

Characteristics of NoSQL Databases

NoSQL database categories

- The most common trait among NoSQL databases is that they are non-relational in architecture
- What types of NoSQL databases are available?
- What is common to them?

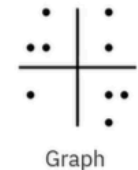


- You've already heard that the most common trait among NoSQL databases is that they are non-relational in architecture. But what types of NoSQL databases are available? And what is common to them?

NoSQL database categories

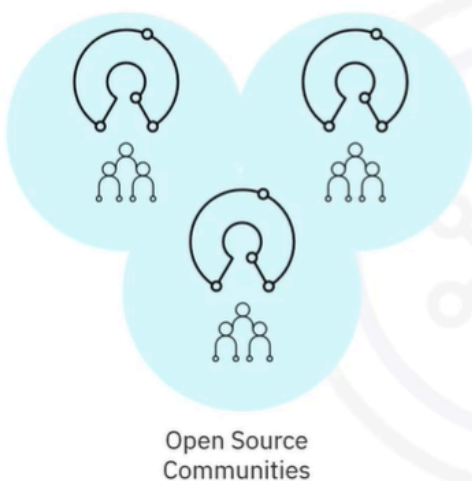
General consensus is...

...NoSQL databases fit into four categories



- Several efforts have been made to categorize NoSQL databases and, in the marketplace, there is a general consensus that they fit into four types: Key-Value, Document, Column-based and Graph style NoSQL databases. There is some overlap among these types, so the definition isn't always clear.

NoSQL database characteristics



But what ties NoSQL databases together?

- Majority have their roots in the open source community
- Many have been used and leveraged in an open source manner
- Open source community support is fundamental to their industry growth

- But what ties NoSQL databases together? One commonality is that the majority of them have their roots in the open-source community and have been used and leveraged in an open-source manner. This has been fundamental for spring-boarding their growth in the industry.

NoSQL database characteristics

- Companies often develop a commercial version of the database alongside the open source version
- Examples include:
 - IBM Cloudant
 - DATASTAX
 - MongoDB



- You'll often see companies who also provide a commercial version of the database, and services and support of the technology, at the same time providing sponsorship of the open-source counterpart. Examples of this include IBM Cloudant for CouchDB, Datastax for Apache Cassandra, and Mongo has their own open source version of the Mongo database too.

NoSQL database characteristics

- Differ technically, but have a few commonalities
- Most NoSQL databases:
 - Are built to scale horizontally
 - Share data more easily than RDBMS
 - Use a global unique key to simplify data sharding
 - Are more use case specific than RDBMS
 - Are more developer friendly than RDBMS
 - Allow more agile development via flexible schemas
- Technically speaking they all differ quite a bit, but a few commonalities do emerge. Most NoSQL databases are built to scale horizontally and share their data more easily than their relational counterparts. To do this often requires the use of a global unique key across a whole database, to simplify partitioning (or 'sharding'). They're also more specialized to certain use cases than RDBMS, which previously have been the Swiss army knives of datastores. Developers are drawn to NoSQL databases for their ease of data modeling and use. And finally, many NoSQL databases allow more agile development through their flexible schemas as compared to the fixed schemas of relational databases.

Benefits of NoSQL databases

Why use a NoSQL database?



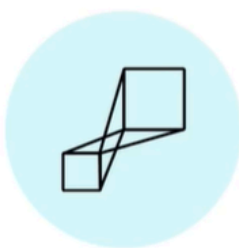
Why is their popularity growing so rapidly?



- But why would you use a NoSQL database? And why is the popularity of these databases growing at such a rapid rate? Well, the list is long, so we've included some of the more common ones here. Let's briefly touch on each topic, keeping in mind that not all NoSQL databases will exhibit all of these benefits.

Benefits of NoSQL databases

Scalability



Server



Server Cluster



Server Racks



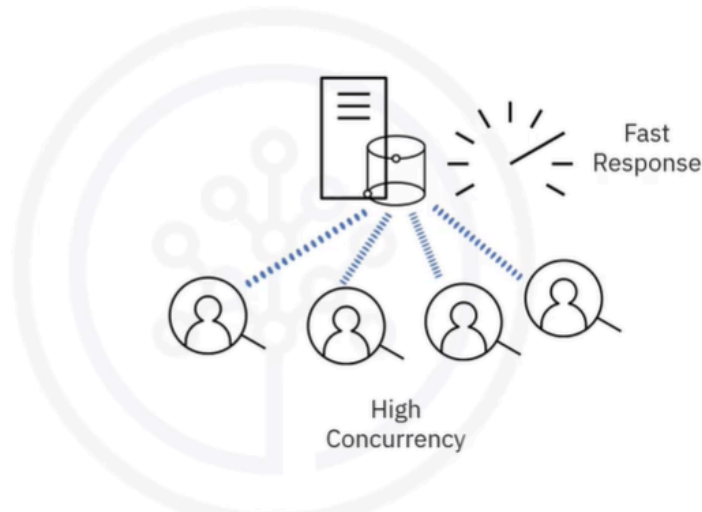
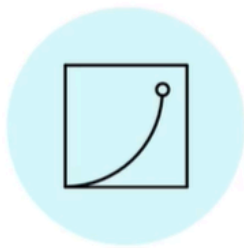
Data Centers

- First, the most common reason to employ a NoSQL database is for scalability, particularly the ability to horizontally scale across clusters of servers, racks,

and possibly even data centers. The elasticity of scaling both up and down to meet the varying demands of applications is key.

Benefits of NoSQL databases

Performance



- The second point of performance goes hand-in-hand with scalability. The need to deliver fast response times even with large data sets and high concurrency is a must for modern applications, and the ability of NoSQL databases to leverage the resources of large clusters of servers makes them ideal for fast performance in these circumstances.

Benefits of NoSQL databases

Availability



Benefits of NoSQL databases

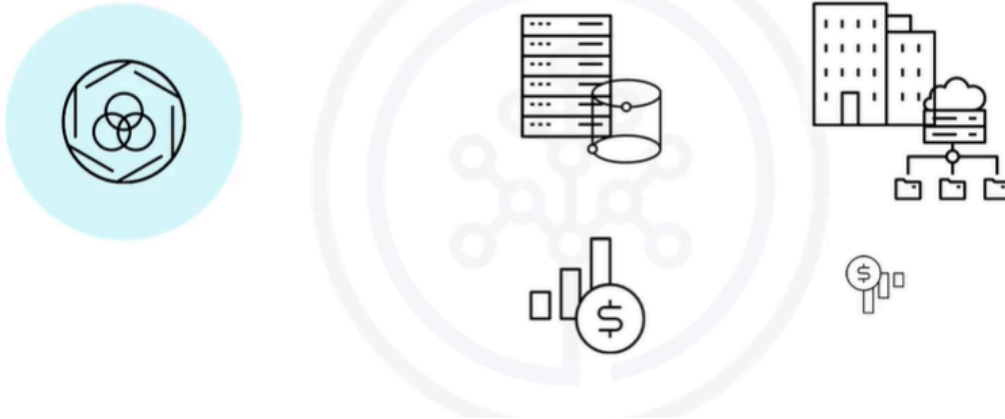
Availability



- High availability is an obvious requirement for a database, and having a database run on a cluster of servers with multiple copies of the data makes for a more resilient solution than a single server solution.

Benefits of NoSQL databases

Cloud Architecture



- Historically, large databases have run on expensive machines or mainframes. Modern enterprises are employing cloud architectures to support their applications, and the distributed data nature of NoSQL databases means that they can be deployed and operated on clusters of servers in cloud architectures, thereby massively reducing cost.

Benefits of NoSQL databases

Cost



- Cost is important for any technology venture, and it is common to hear of NoSQL adopters cutting significant costs versus their existing databases... and still be able to get the same or better performance and functionality.

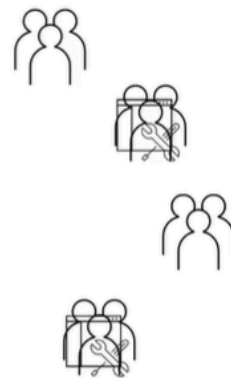
Benefits of NoSQL databases

Flexible Schema



Benefits of NoSQL databases

Flexible Schema



- Flexible schema and intuitive data structures are key features that developers love when wanting to build applications efficiently. Most NoSQL databases allow for having flexible schemas, which means that one can build new

features into applications quickly and without any database locking or downtime.

Benefits of NoSQL databases

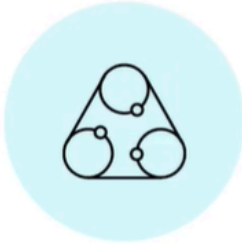
Varied Data Structures



- NoSQL databases also have varied data structures, which often are more eloquent for solving development needs than the rows and columns of relational datastores. Examples include key-value stores for quick lookup, document stores for storing de-normalized intuitive information, and graph databases for associative data sets.

Benefits of NoSQL Databases

Specialized capabilities

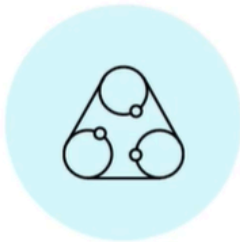


Indexing and Querying

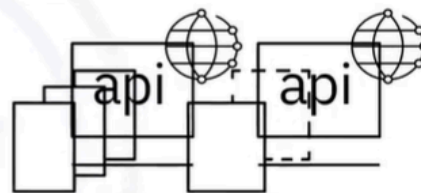


Benefits of NoSQL Databases

Specialized capabilities



Indexing and Querying

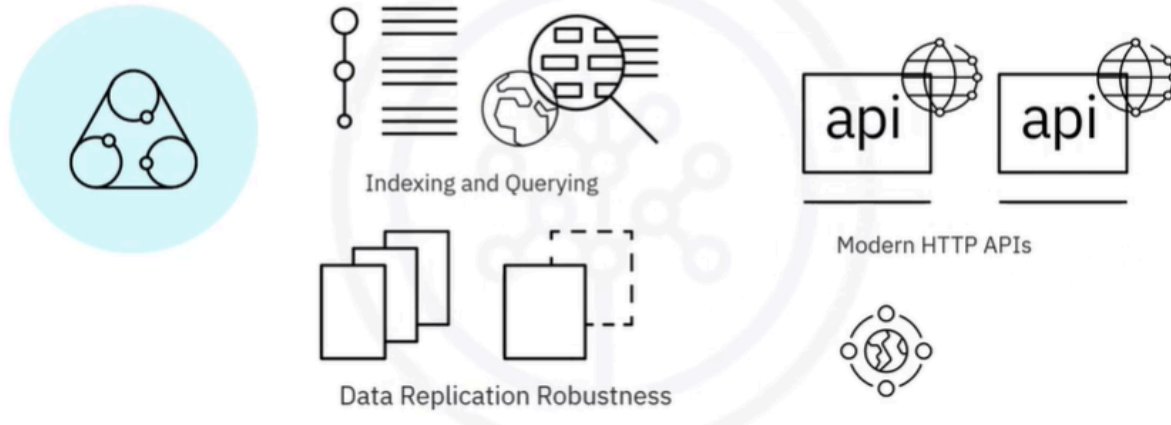


Modern HTTP APIs
Data Replication Robustness



Benefits of NoSQL Databases

Specialized capabilities



- There are also various specialized capabilities that certain NoSQL providers offer that attract end users. Examples include specific indexing and querying capabilities such as geospatial search, data replication robustness, and modern HTTP API's. With all these benefits you might well ask why anyone would ever use anything but a NoSQL database? Well, you could say this is true for most cases these days, but there are definitely still some requirements which are best met with an RDBMS.

Recap

In this video, you learned that:

- NoSQL databases are non-relational
- There are four categories of NoSQL databases
- NoSQL databases have their roots in the open-source community
- NoSQL database implementations are technically different
- There are several benefits to adopting NoSQL databases

